

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEEOE605B - POWER PLANT ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 A plant delivers 500 kW for 12 hours and 1,000 kW for the next 12 hours. What is the average load?
 - A. 600 kW
 - B. 700 kW
 - C. 750 kW
 - D. 1,500 kW
- In an economic analysis of a plant, which of the following is considered a direct cost?
 - A. Taxes and insurance
 - B. Administrative salaries
 - C. Raw material cost
 - D. Building rent
- 3 A moderator, in nuclear power plants, is a medium introduced into the fuel mass in order to _____
 - A. Control the reaction
 - B. Reduce the temperature
 - C. Extract heat from nuclear reaction
 - D. Slow down the speed of fast moving neutrons
- 4 Which type of turbine is used for high head and low discharge conditions?
 - A. Kaplan turbine
 - B. Pelton turbine
 - C. Francis turbine
 - D. Crossflow turbine
- 5 Which plant has the fastest response time to sudden load changes in a combined system?
 - A. Steam turbine plant
 - B. Hydro power plant
 - C. Nuclear power plant
 - D. Thermal power plant

- 6 Which of the following is a major technical challenge in MHD power generation?
- Low working temperatures
 - Lack of available magnetic materials
 - Corrosion of electrode materials
 - Inability to ionize gases

- 7 The cut-in speed of a wind turbine refers to:
- The speed at which the turbine begins generating usable power
 - The speed at which the turbine shuts down for safety
 - The maximum rated speed of the turbine
 - The wind speed at which noise is minimum

Modern steam power plants work on which of the following cycle?

- Carnot cycle
- Rankine cycle
- Otto cycle
- Bell- Coleman cycle

- 9 The path of flue gases in Thermal power plant is
- Boiler – Economizer – Superheater– Air preheater
 - Boiler – Superheater – Air preheater – Economizer
 - Boiler – Air preheater – Superheater – Economizer
 - Boiler – Superheater – Economizer – Air preheater

The load factor is _____.

- always less than unity
- less than or greater than 1
- always greater than 1
- less than zero

A base load plant typically operates

- Only during peak hours
- Intermittently
- Continuously throughout the year
- Only during maintenance periods

- 12 A domestic consumer has a flat tariff of ₹4.50 per unit. If the consumer uses 12 units per day, what is the monthly (30 days) bill?

- ₹2,100
- ₹1,620
- ₹2,250
- ₹2,340

Q2. Solve the following.

- A) Compare thermal, hydro and nuclear power plant on the basis of different parameters.

- B) Generating stations has the following daily load cycle:

Time (Hours)	0 - 6	6 - 12	12 - 14	14 - 16	16 - 20	20 - 24
Load (MW)	45	55	60	50	80	45

Draw load curve and load duration curve and find

- Maximum demand
- Units generated per day
- Average load
- Load factor

- Q3. Solve the following.
- Explain the operational circuits of thermal power plant with Schematic diagram of thermal power plant. 6
 - Classify the different types of Hydropower plant. Explain Pumped Storage Plant with its layout. 6
- Q4. Solve Any Two of the following.
- List the types of reactors in nuclear power plant and explain PWR reactor with its layout. 6
 - Describe the components of gas power plant with its schematic diagram and also state the advantages and disadvantages of gas power plant. 6
 - State and explain types of diesel power plants. 6
- Q5. Solve Any Two of the following.
- Explain the working principle of photovoltaic solar cells. Discuss types of PV materials. 6
 - With the help of neat diagram explain power generation from geothermal power plant. 6
 - Explain with a neat diagram the working principle and components of Magneto-hydrodynamic (MHD) power generation. 6
- Q6. Solve Any Two of the following.
- Explain concept of grid and conditions to interface different power plant to grid. 6
 - What are the different types of costs associated with electricity generation. 6
 - The daily load of an industrial consumer is as follows: 200 kW for 5 hours, 100 kW for 12 hours, 80 kW for 7 hours and 5 kW for 2 hours. The tariff rate is Rs. 500 per kW of maximum demand per year plus Rs. 2.50 per kWh. Determine the energy consumption per year and yearly bill. 6

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